

Spatial Economics - TA Session 1

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Canonical QSM from Redding and Rossi-Hansberg (2017)

- Locations $n \in N$, differentiated firms $j \in [0, M_n]$, representative households
- Preferences over consumption bundle C_n and residential land use h_n

$$U_n = \left(\frac{C_n}{\alpha} \right)^\alpha \left(\frac{h_n}{1-\alpha} \right)^{1-\alpha}, \quad 0 < \alpha < 1$$

where

$$C_n = \left[\sum_{i \in N} \int_0^{M_i} c_{ni}(j)^{\frac{\sigma-1}{\sigma}} dj \right]^{\frac{\sigma}{\sigma-1}}, \quad P_n = \left[\sum_{i \in N} \int_0^{M_i} p_{ni}(j)^{1-\sigma} dj \right]^{\frac{1}{1-\sigma}}$$

- Linear technology with fixed cost: producing $x_i(j)$ requires $l_i(j) = F + x_i(j)/A_i$

Model Inversion - Example

Consider an economy with

- parameters $\Sigma = \{\sigma, \alpha, \{d_{ni}\}_{i \in N, n \in N}\}$
- (exogenous) fundamentals $A = \{A_i, H_i\}_{i \in N}$
- (endogenous) allocations $Y = \{L_i\}_{i \in N}$
- (endogenous) prices $P = \{w_i\}_{i \in N}$
- agents' optimality conditions and market clearing $F(\Sigma, A, Y, P) = 0$
- equilibrium defined as:

Competitive Equilibrium (CE)

Given parameters and fundamentals $\{\Sigma, A\}$, a CE is a set of prices and allocations $\{P, Y\}$ such that given prices P , allocations Y solve $F(\Sigma, A, Y, P) = 0$.

Model Inversion - Roadmap

- Ways to set parameters
 - externally calibrate: borrow from a different paper
 - internally calibrate: use data to estimate it in a model-consistent way (SMM, IV)
 - simulate: set different values on a grid and see how model behaves
- We usually observe allocations and prices $\{Y, P\}$
- In general, we do not observe fundamentals A

Model Inversion - Framework

- Model: we assume the data we observe is generated by:

$$\Sigma + \underbrace{A}_{\text{unobserved}} \xrightarrow{F} \underbrace{Y, P}_{\text{observed}}$$

- Inversion: find A that rationalizes data under the assumptions of the model

$$\Sigma + \underbrace{Y, P}_{\text{observed}} \xrightarrow{F} \underbrace{A}_{\text{unobserved}}$$

- Counterfactuals: ask how changes in exogenous objects would change the outcome

$$\underbrace{\hat{\Sigma} + \hat{A}}_{\text{assumed}} \xrightarrow{F} \underbrace{\hat{Y}, \hat{P}}_{\text{object of interest}}$$

Model Inversion - Caveats

- Inverting the model requires a one to one mapping between Σ, Y, P and A
- Otherwise the mapping is over/under-identified and may not be very useful
- A stylized example. If for $|N|$ regions
 - the competitive equilibrium is unique
 - we observe $2|N|$ variables $\{L_i, w_i\}_{i \in N}$
 - F consists of $2|N|$ equations
 - fundamentals consists of $2|N|$ variables $\{A_i, H_i\}_{i \in N}$then, we can recover $\{A_i, H_i\}_{i \in N}$ that rationalizes the data under model assumptions
- We may not observe all of Y, P or hesitate to use it
- Model assumptions are a big degree of freedom
- Whether to invert the model depends on our objective, data, and assumptions

Model Inversion - Application to Redding and Rossi-Hansberg (2017)

- I want to find $\{A_i, H_i\}_{i \in N}$ that rationalizes the model
- I have data for $\{L_i, w_i\}_{i \in N}$
- I externally calibrate $\sigma = 5$, $\alpha = 0.75$, and $\{d_{ni}\}_{i \in N, n \in N}$ based on distances
- I solve the optimality and market clearing conditions, $\forall n \in N$:

$$L_n^{\tilde{\sigma}\gamma_1} A_n^{-\frac{(\sigma-1)(\sigma-1)}{2\sigma-1}} H_n^{-\frac{\sigma(\sigma-1)(1-\alpha)}{\alpha(2\sigma-1)}} \\ = \bar{W}^{1-\sigma} \sum_{i \in N} \frac{1}{\sigma F} \left(\frac{\sigma}{\sigma-1} d_{ni} \right)^{1-\sigma} \left(L_i^{\tilde{\sigma}\gamma_1} \right)^{\frac{\gamma_2}{\gamma_1}} A_i^{\frac{\sigma(\sigma-1)}{2\sigma-1}} H_i^{\frac{(\sigma-1)(\sigma-1)(1-\alpha)}{\alpha(2\sigma-1)}}$$

and

$$\tilde{\zeta} = w_n^{1-2\sigma} A_n^{\sigma-1} L_n^{(\sigma-1)\frac{1-\alpha}{\alpha}} H_n^{-(\sigma-1)\frac{1-\alpha}{\alpha}}$$

Redding, Stephen J and Esteban Rossi-Hansberg, “Quantitative spatial economics,” *Annual Review of Economics*, 2017, 9 (1), 21–58.